Fixed Income Analysis Report

GBP-Collateralized FX Forward and Interest-Rate Products

Currency-Chain Discounting, Fair Delivery Price, RFR Swaps and Overnight Basis Swaps

This report derives the value and fair strike of an EUR/USD forward collateralized in GBP, establishes when the supplied discount factors are sufficient through a USD-centered currency chain, and documents the completed implementation of RFR swaps, overnight-index basis swaps, convention-driven product factories and composite valuation engines.

Author Ziyi Yang

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Abstract

An FX forward is the difference between two deterministic currency claims. Under GBP collateralization, the EUR claim and the USD claim must both be valued with GBP-collateralized effective discount factors. Their ratio gives the fair forward strike. Under the course assumption that an FX forward is independent of collateral currency, a USD-centered currency chain constructs both required curves from the supplied SONIA, SOFR and USD-collateralized cross-currency factors. The programming component builds and values fixed- versus-overnight swaps and SOFR-versus-Fed-Funds basis swaps.

Keywords: FX forward • Foreign collateral • Currency chain • RFR swap • Basis swap

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1 Setup and Notation

Let $S_t^{\$/\mathsf{EUR}}$ be the spot exchange rate in USD per EUR. A long forward with EUR notional N receives N EUR and pays NK USD at maturity T. Let $P^{x,c}(t,T)$ denote the time-t value, in currency x, of one unit of currency x paid at T when the collateral currency is c.

The position consists of two deterministic claims:

- receive one EUR at T, whose USD value today is $S_t^{\$/\mathsf{EUR}} P^{\mathsf{EUR},\mathsf{GBP}}(t,T)$;
- pay K USD at T, whose USD value today is $K P^{\$,\mathsf{GBP}}(t,T)$.

2 Q1: Valuation Formula

Question focus. *Value the EUR/USD forward in USD when the CSA collateral currency is GBP.*

Combining the two claims gives

$$\boxed{V_t^{\$} = N\left[S_t^{\$/\mathsf{EUR}} P^{\mathsf{EUR},\mathsf{GBP}}(t,T) - K P^{\$,\mathsf{GBP}}(t,T)\right].} \tag{2.1}$$

Define the GBP-collateralized forward rate by

$$F_t^{\$/\mathsf{EUR};\mathsf{GBP}}(T) = S_t^{\$/\mathsf{EUR}} \frac{P^{\mathsf{EUR},\mathsf{GBP}}(t,T)}{P^{\$,\mathsf{GBP}}(t,T)}. \tag{2.2}$$

Then the mark-to-market takes the familiar form

$$\boxed{V_t^{\$} = N P^{\$,\mathsf{GBP}}(t,T)\left[F_t^{\$/\mathsf{EUR};\mathsf{GBP}}(T) - K\right].} \tag{2.3}$$

The collateral currency appears in both discount factors. Replacing either one with an ordinary domestic curve would mix inconsistent collateral regimes.

3 Q2: Fair Forward Strike

At inception, the fair strike sets the forward value to zero. Hence

$$S_t^{\$/\mathsf{EUR}} P^{\mathsf{EUR},\mathsf{GBP}}(t,T) = K_{\mathsf{fair}} P^{\$,\mathsf{GBP}}(t,T), \tag{3.1}$$

and therefore

$$\boxed{K_{\mathsf{fair}} = S_t^{\$/\mathsf{EUR}} \frac{P^{\mathsf{EUR},\mathsf{GBP}}(t,T)}{P^{\$,\mathsf{GBP}}(t,T)}.} \tag{3.2}$$

The numerator discounts the received base-currency claim and the denominator discounts the paid quote-currency claim under the same GBP collateral rule.

4 Q3: Sufficiency of the Supplied Curves

The curve set provides

$$P^{\mathsf{GBP},\mathsf{GBP}} = df_{\mathsf{SONIA}}, \quad P^{\mathsf{EUR},\mathsf{EUR}} = df_{\mathsf{EuroSTR}}, \quad P^{\$,\$} = df_{\mathsf{SOFR}}, \tag{4.1}$$

together with $P^{\mathsf{GBP},\$} = df[\mathsf{GBP}, \$]$ and $P^{\mathsf{EUR},\$} = df[\mathsf{EUR}, \$]$.

Under the course currency-chain assumption that the same-maturity FX forward is independent of the collateral currency, a USD-centered chain gives

$$P^{\mathsf{EUR},\mathsf{GBP}} = \boxed{\frac{P^{\mathsf{EUR},\$}}{P^{\mathsf{GBP},\$}} P^{\mathsf{GBP},\mathsf{GBP}}}, \tag{4.2}$$

$$P^{\$,\mathsf{GBP}} = \boxed{\frac{P^{\$,\$}}{P^{\mathsf{GBP},\$}} P^{\mathsf{GBP},\mathsf{GBP}}}. \tag{4.3}$$

Both GBP-collateralized factors required by the valuation formula can therefore be constructed from the supplied set. In the fair-strike ratio, the GBP terms and the GBP/USD cross-currency factor cancel:

$$\boxed{K_{\mathsf{fair}} = S_t^{\$/\mathsf{EUR}} \frac{P^{\mathsf{EUR},\$}}{P^{\$,\$}}.} \tag{4.4}$$

Key result. The supplied curves are sufficient under the assignment's currency-chain assumption. The EUR domestic df_{EuroSTR} is not required for this particular forward once $P^{\mathsf{EUR},\$}$ is available.

The conclusion is conditional. If forward prices can depend materially on the collateral currency because of CSA optionality, liquidity, transaction costs or convexity effects, the chain identity is not automatic. Then direct $P^{\mathsf{EUR},\mathsf{GBP}}$ and $P^{\$,\mathsf{GBP}}$ curves, or equivalent GBP-collateralized cross-currency basis data, are needed.

5 Completed Product Library

5.1 RFR swap

The completed product constructor builds a fixed stream and a compounded overnight stream with their contractual accrual periods and payment conventions. For receive-fixed,

$$PV_{\mathsf{swap}} = PV_{\mathsf{fixed}} - PV_{\mathsf{floating}}. \tag{5.1}$$

The floating coupons use the SOFR projection curve, while both legs are discounted with the selected USD funding curve. The same signs are applied to cash paid or received on the value date.

5.2 Overnight-index basis swap

Leg 1 contains compounded SOFR plus the quoted spread; leg 2 contains compounded Fed Funds. Each index uses its own projection curve. The implementation also exposes the leg-1 index and spread as separate streams, so the valuation engine combines

$$PV_{\mathsf{basis}} = PV_{\mathsf{leg1,index}} + PV_{\mathsf{leg1,spread}} - PV_{\mathsf{leg2,index}} \tag{5.2}$$

when leg 1 is received.

5.3 Convention-driven factory

The two completed builders map stored conventions into product constructors. For swaps, `3Mx1Y` means a start three months after the value date and a one-year life from that start; `1Y` means spot-starting for one year. The quote is the fixed rate for an RFR swap and the spread over leg 1 for a basis swap. The notebook verifies that factory-built products serialize identically to directly built products.

Table 1. Verified Assignment 6 product values

Product	PV	Verification
FRA	87.090294	Matches the analytical forward-rate formula
Receive-fixed RFR swap	$-2{,}629.679575$	Fixed-leg PV minus floating-leg PV
Receive-leg-1 basis swap	$8{,}446.472471$	SOFR-plus-spread PV minus Fed-Funds PV

The executed notebook additionally checks a known FRA fixing, cashflow reports, factory equality and the public valuation API. All numerical assertions pass.

6 Conclusion

Foreign-collateral FX pricing requires one effective discount factor for each settlement currency under the common CSA currency. The fair strike is their ratio times spot. Under the stated currency-chain assumption, the supplied USD-centered curves are sufficient to reconstruct both GBP-collateralized factors. The completed code applies the same decomposition principle to linear rate products: construct contractual cashflow streams, project each floating leg with its index curve, discount with the funding curve and combine the legs with the correct receive/pay signs.